

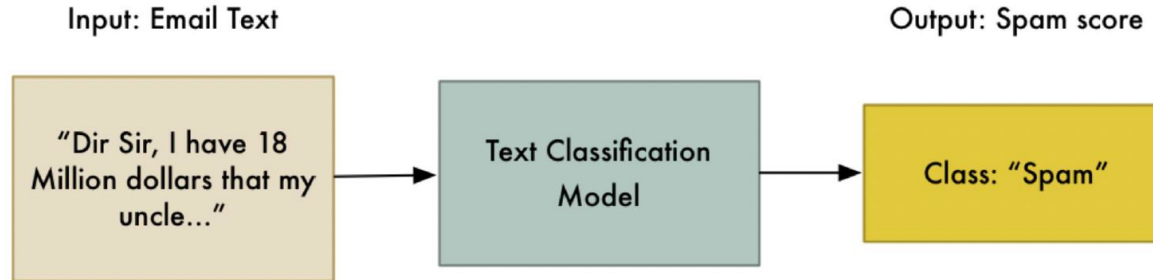


MASTER OF SCIENCE
IN ENGINEERING

Text Classification and Sentiment Analysis

Text classification = assigning text documents to one or more categories

- We have a predefined set of classes
- We want to assign a previously unseen document to one of such classes (label it)
- If we have 2 classes, it's binary classification (e.g.: spam vs. no spam)
- We may have more than 2 classes (e.g.: politics, finance, sports, . . .)
- There exist specific forms of text classification
 - sentiment analysis: the labels are positive, negative, neutral



Source: Adam Geitgey, medium.com

Type	Definition
Sentiment analysis	Classification based on the polarity (positive, negative, or neutral) or emotional tone of text.
News classification	Categorize news articles based on content and subject matter.
Topic labeling	Identify the main subject or topic in text.
Question-answering	Provide accurate answers to user queries.
Natural language inference	Determine the logical relationship (entailment, contradiction, or neutral) between sentences.
Dialog act classification	Identification of text intention such as requesting, questioning, or informing.
Named entity recognition	Categorize named entities such as people, organizations, and locations.
Syntactic parsing	Analyze the grammatical structure of text to understand the relationship between words and phrases.

Source: John Fields et al., A Survey of Text Classification With Transformers: How Wide? How Large? How Long? How Accurate? How Expensive? How Safe?
IEEE Access, 2023

Some examples of data sources:

- Social Media
- Surveys
- Web Scraping
- Emails
- User Reviews
- Comments

Cleaning up the data is a very complex task



Binary Classification

- Assigns one of two mutually exclusive categories to a text.
- Example: Spam detection where emails are classified as 'Spam' or 'Not Spam'.
- Common in tasks with clear opposites or binary outcomes.

Multiclass Classification

- Assigns one of three or more mutually exclusive categories to a text.
- Example: Classifying a text into genres such as 'Fiction', 'Non-Fiction', or 'Poetry'.
- Useful for organizing data into distinct, non-overlapping groups.

Fine-Grained Sentiment Analysis

- Common example of a multiclass
- Captures more nuances
 - Positive vs negative vs neutral
 - Amazon's five star rating system
- Easy to go overboard

Aspect-Based Sentiment Analysis

- Analyzes sentiments toward specific aspects or features of a product.
- Example: Evaluating a product's quality, price, or design separately.

Offensive Language Detection

- Identifies text containing abusive, offensive, or inappropriate content.
- Common in moderating social media or public forums.

Multilabel Classification

- Assigns multiple overlapping labels to a single text.
- Example: Labeling a comment for both 'Violence' and 'Hate Speech'.
- Suitable for complex tasks where texts can belong to multiple categories.

Intent Recognition

- Trains models on dialog to understand user intent.
- Example: Recognizing intents like 'Buy', 'Query', or 'Complain'.

Document classification

- Classify longer documents like articles, research papers, books...
- Traditionally done using standard non-neural techniques (still very useful as a baseline)
- Shifted toward word embeddings (circa 2015) and contextual embeddings (circa 2018)
- Today, LLM-based

Fake News Classification

- Identifies patterns to classify news as true or fake.
- Useful for combating misinformation in media.
- Incredibly hard to do

Cross-Lingual Classification

- Different from language detection, which is relatively simple (depending on the language)
- Classification of multi-lingual content into language-independent categories
- Example: multilingual reviews on TripAdvisor

Stance Detection

- Determines the perspective toward a piece of text (in favor, against, neutral)
- Applicable e.g. to political debates, social issues, ...
- Conceptually similar to sentiment analysis but applied to text of different nature

Emotion/Mental Health Detection

- Assesses the emotional or mental health status of a person from text.
- Detecting stress, anxiety, or depression in online interactions.

Multimodal Classification

- Combines multiple data types such as text, video, and audio for classification
- Example: Analyzing video subtitles along with tone and gestures

Performance Evaluation

There is a standard way to evaluate the performance of any classifier.

- Accuracy: fraction of test documents that have been correctly classified
- Precision P_c for class c : fraction of test documents *classified as c* that have been correctly classified
- Recall R_c for class c : fraction of test documents *labeled as c* that have been correctly classified
- F_1 score for class c : harmonic mean of precision and recall

$$F_1 = \frac{2P_c R_c}{P_c + R_c} \quad (1)$$

Performance Evaluation

Given a class c :

- True Positive (TP): you see a c sample and classify it correctly
- False Positive (FP): you see a non- c sample and classify it incorrectly as c
- True Negative (TN): you see a non- c sample and classify it correctly
- False Negative (FN): you see a c sample and classify it incorrectly as non- c

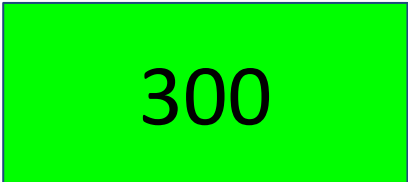
$$P_c = \frac{TP}{TP + FP}$$

$$R_c = \frac{TP}{TP + FN}$$

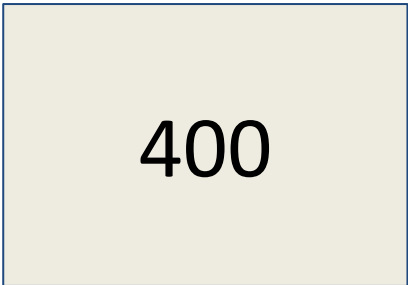
Example



300



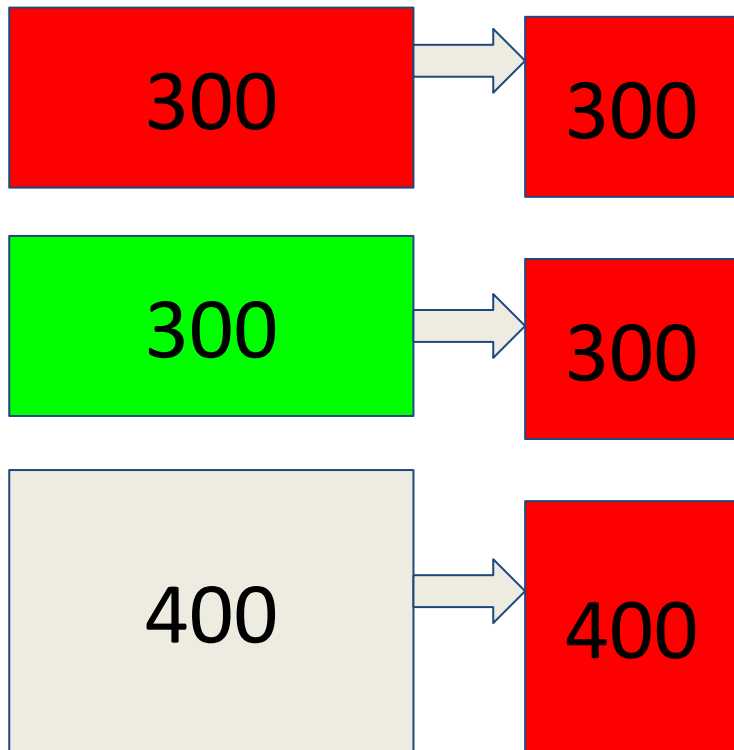
300



400

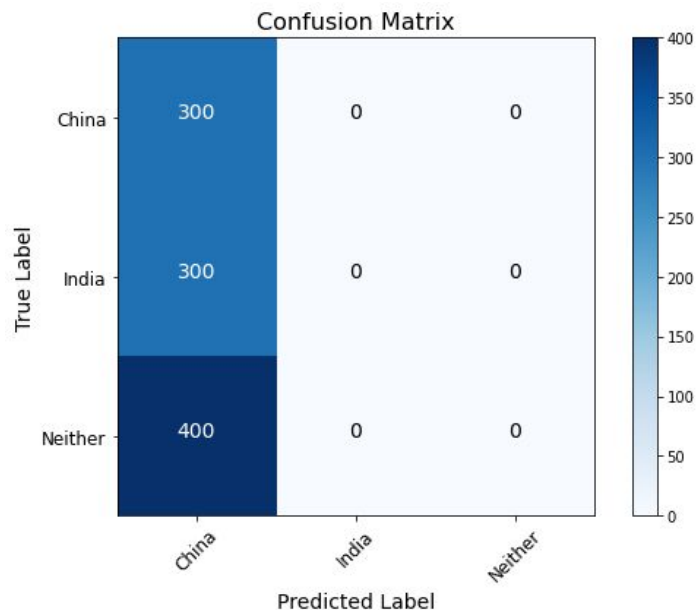
- Imagine we build a 3-way classifier for news articles
- Classes: China, India, neither
- We have 300 China articles, 300 India article, and 300 “neither” articles

Dumb classifier



- Accuracy: 0.3
 - 300 correct decisions out of 1000
- Precision for China: 0.3
 - 300 TP out of (300 TP + 700 FP)
- Recall for China: 1
 - 300 TP out of (300 TP + 0 FN)
- Precision for India: N/A
 - No India decisions!
- Recall for India: 0
 - 0 TP out of (0 TP + 300 FN)
- Precision for Neither: N/A
- Recall for Neither: 0
- F1 for China: 0.46
- F1 for India: N/A

Dumb classifier



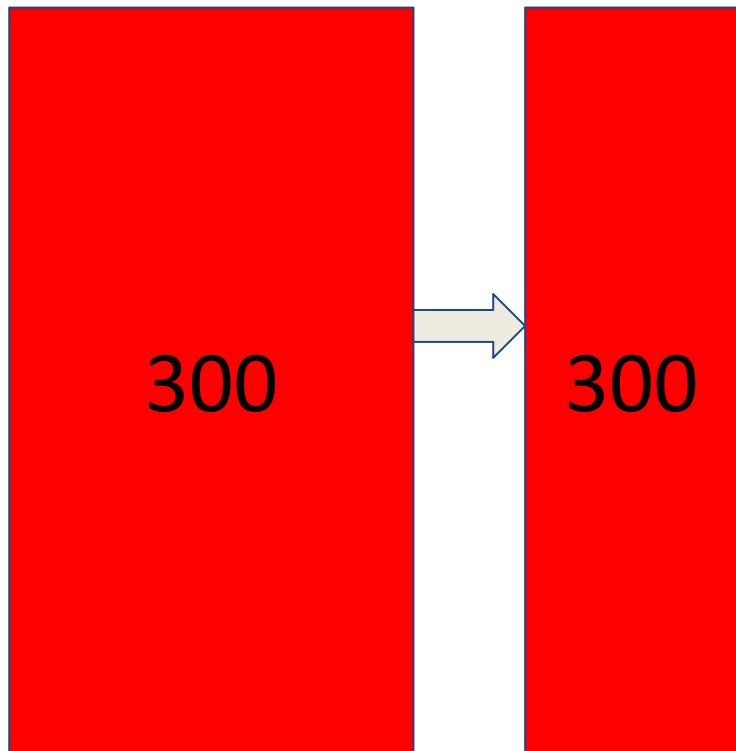
- Precision for India: N/A
 - We get 0/0
- Recall for India: 0
 - We get 0/300

- Accuracy: $\text{trace}(\text{sum of elements along the main diagonal}) / \text{grand sum (sum of all elements)}$
- Precision: given a column (i.e., a predicted label), $\text{element on main diagonal} / \text{sum of matrix column}$
- Recall: given a row (i.e., a test label), $\text{element on main diagonal} / \text{sum of matrix row}$

Dumb classifier

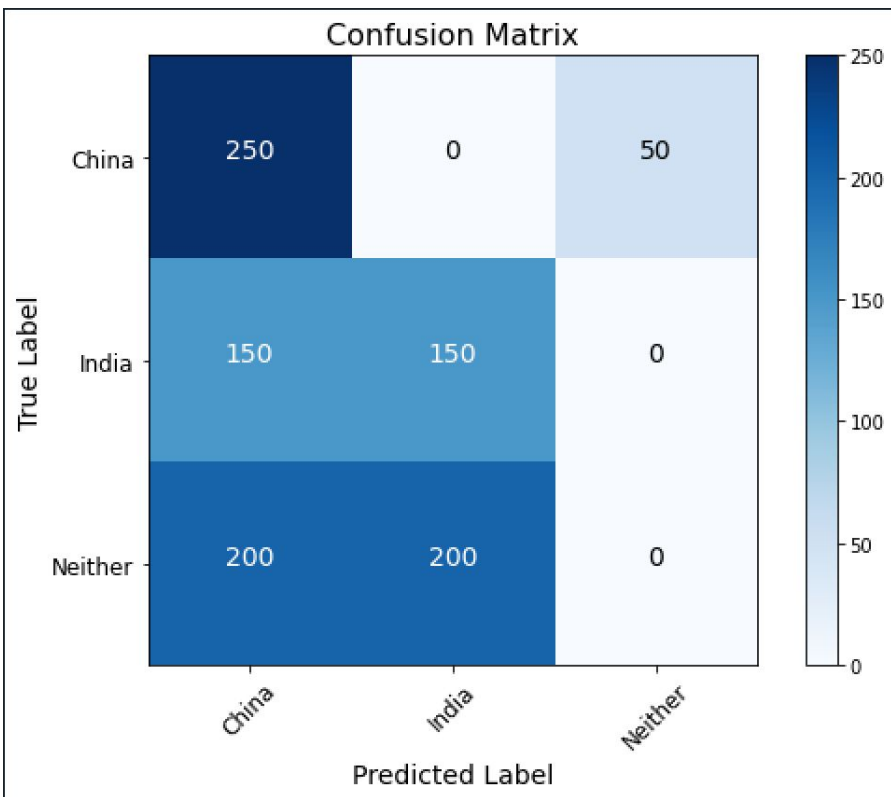
	precision	recall	f1-score	support
China	0.30	1.00	0.46	300
India	0.00	0.00	0.00	300
Neither	0.00	0.00	0.00	400
accuracy	0.30	0.30	0.30	0
macro avg	0.10	0.33	0.15	1000
weighted avg	0.09	0.30	0.14	1000

Dumb classifier looking good?



- Accuracy: 1
- Precision for China: 1
- Recall for China: 1
- Precision for India: N/A
- Recall for India: N/A
- Precision for Neither: N/A
- Recall for Neither: N/A
- F1 for China: 1

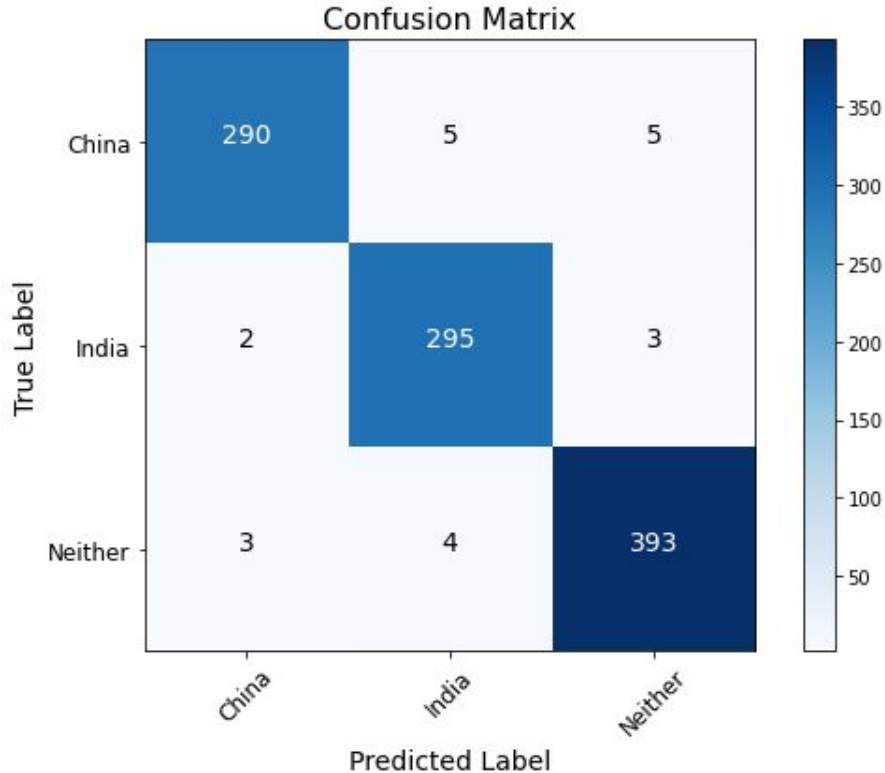
A better classifier



	precision	recall	f1-score	support
China	0.42	0.83	0.56	300
India	0.43	0.50	0.46	300
Neither	0.00	0.00	0.00	400
accuracy	0.40	0.40	0.40	0
macro avg	0.28	0.44	0.34	1000
weighted avg	0.25	0.40	0.31	1000

Still very bad, but
relatively better
than the dumb one

A good classifier



	precision	recall	f1-score	support
China	0.98	0.97	0.97	300
India	0.97	0.98	0.98	300
Neither	0.98	0.98	0.98	400
accuracy	0.98	0.98	0.98	0
macro avg	0.98	0.98	0.98	1000
weighted avg	0.98	0.98	0.98	1000

Having big
numbers on
the diagonal
bodes well



How do we build it?

- Not so fast; first go collect some data!
- Get it cleaned up and properly labeled
- Got clean labeled data? You've done most of the work
- Turn the text data into numbers
- Now you can choose one of a gazillion ways to classify your data
- Broadly speaking, you have three options:
 - Rule-based hand-crafted fully-manual approach
 - Supervised ML (legacy ML)
 - Bleeding-edge NLP

Legacy ML

- Many well-established, easy-to-use algorithms
- Near zero coding effort for anyone
- No need to know what's going on under the hood



Well... it does help if you understand what's going on under the hood.



Sure, but life is short...

Legacy ML classifiers

How to use them:

- Transform each training example into a vector
- Use the vectorized training set to train some ML model

Which ML model? Take your pick!

- Naive Bayes
- Logistic Regression
- Support Vector Machines
- kNN
- Decision Trees
- Random Forests
- Gradient Boosting Machines

Logistic Regression in a nutshell



- Predicts probabilities for binary classes (e.g., positive vs negative sentiment)

1. Linear Model:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

2. Sigmoid Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Converts z into a probability between 0 and 1.

- ## 3. Decision Boundary:
- If $P(y = 1|x) \geq 0.5$, predict the positive class. - Otherwise, predict the negative class.



What if I have more than 2 classes?

Logistic Regression

- ▶ **Binary Classification:** - Logistic regression uses a single sigmoid function.
- ▶ **Multiclass Classification:** - For more than two classes, it uses the **softmax function** to predict probabilities for each class.

$$\text{Softmax: } P(y = i|x) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

The probabilities must add up to 1, hence the need for a softmax

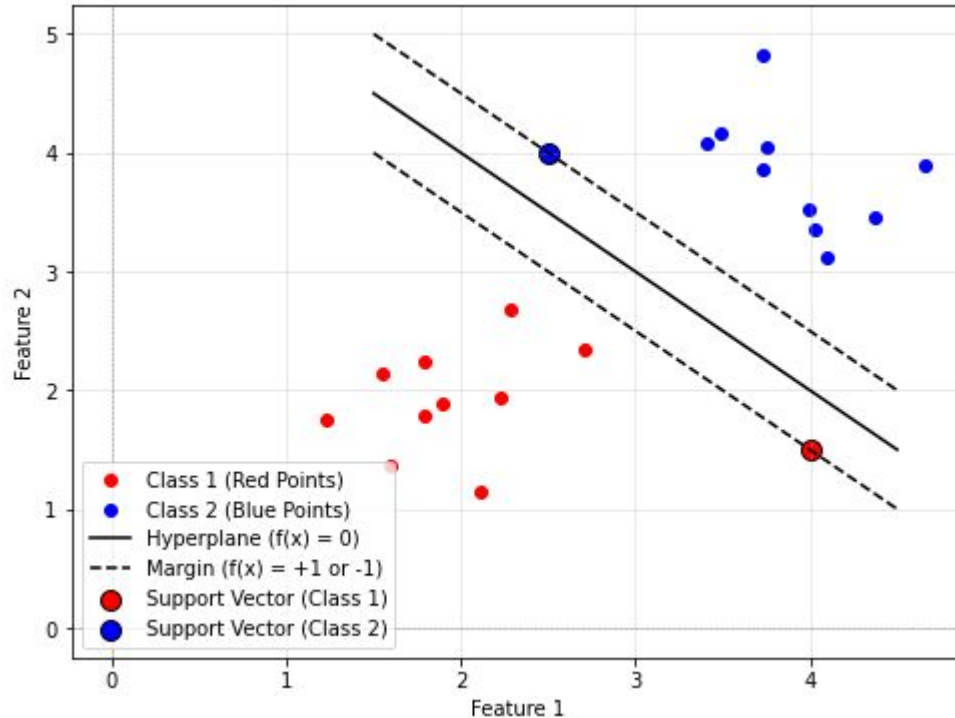


With a softmax you go from a bunch of arbitrary real numbers to a bunch of real numbers that add up to 1!

Support Vector Machines

- ▶ A **Support Vector Machine (SVM)** is a supervised machine learning algorithm used for:
 - ▶ **Classification:** Binary or multiclass classification tasks.
 - ▶ **Regression:** Predicting continuous values.
- ▶ SVMs aim to find the **optimal hyperplane** that separates data points belonging to different classes.
- ▶ The decision boundary is chosen to maximize the **margin** between the classes.

Support Vector Machines



- The **hyperplane** separates the two classes.
- The **support vectors** are the closest points to the hyperplane, lying on the **margin boundaries** (dashed lines).
- The **margin size** is determined by the perpendicular distance between the hyperplane and these margin boundaries.

SVM hyperplane

- ▶ A hyperplane in SVM is defined as:

$$w \cdot x + b = 0$$

- ▶ Here:
 - ▶ w : The normal vector (perpendicular to the hyperplane).
 - ▶ x : The feature vector.
 - ▶ b : The bias term.
- ▶ The hyperplane separates the data into two classes:

$$w \cdot x + b > 0 \quad (\text{Class 1}) \quad \text{and} \quad w \cdot x + b < 0 \quad (\text{Class 2}).$$

Optimal hyperplane

In **SVM**, we find an **optimal hyperplane** such that:

- **Support vectors** lie **exactly** on the **margin boundaries**.
- The **margin boundaries** are **parallel to the optimal separating hyperplane**.
 - Positive hyperplane, where the support vectors of the positive class lie
 - Negative hyperplane, where the support vectors of the negative class lie
- The margin (distance between the boundaries) is **maximized**.



[Original
paper](#)

Machine Learning, 20, 273–297 (1995)
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Support-Vector Networks

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From legacy ML to Transformers

- Legacy ML methods require preprocessing: features must be explicitly pre-computed and aren't learned automatically at training time
- RNNs learn representations dynamically, but operate sequentially and have problems with long-range dependencies
- Transformers learn representations dynamically and leverage the attention mechanism to move past the limitations of RNNs

Sentiment Analysis

Which of these sentences express sentiment, and what is their sentiment polarity?

1. There was an earthquake in California.
2. The team failed to complete the physical challenge. (We win/lose!)
3. They said it would be great.
4. They said it would be great, and they were right.
5. They said it would be great, and they were wrong.
6. The party fat-cats are sipping their expensive imported wines.
7. Oh, you're terrible!
8. Here's to ya, ya bastard!
9. Of 2001, "Many consider the masterpiece bewildering, boring, slow-moving or annoying, . . ."
10. long-suffering fans, bittersweet memories, hilariously embarrassing moments, . . .

Source: Christopher Potts, CS224U 2019, Stanford University

Sentiment Analysis

Understanding the emotional state of the author of a text

- also called opinion mining
- very lucrative applications (marketing, voice of the customer)
- can be extremely hard

In principle, it's very different from text classification

- Text classification: what's this text about?
- Sentiment analysis: how does the author of this text feel about the subject of this text?

But there are striking similarities

- The text can reflect a positive or a negative emotional state

Application Examples

It can be applied to anything that contains subjective opinions

- Movie reviews: is this review **negative** or **positive**?
- Product reviews: how do people feel about a new product?
- Stock markets: what's the general market sentiment?
- Politics: how does the public feel about the candidates and their agendas?

Sentiment analysis is often carried out with **prediction** in mind

Formal definition

Commonly framed as **attitude detection**

- Attitude: enduring disposition toward someone or something with an emotional connotation
- The attitude can be captured as a quintuple
 - ① opinion holder: whose opinion is this?
 - ② target entity: what is this opinion about?
 - ③ aspect: specific feature of the target that this opinion is about
 - ④ type: most commonly **positive**, **negative**, **neutral**, along with an indication of strength
 - ⑤ time when this opinion was expressed
- Opinion target = target entity + aspect
- Knowing the opinion without knowing the opinion target is of limited use

Example

Our cozy room at the Overlook
Hotel had a great view.

J. Nicholson, Neptune City, New
Jersey, USA

- ① opinion holder: J. Nicholson
- ② target entity: the Overlook Hotel
- ③ aspect: J. Nicholson's room
- ④ type: **positive**
- ⑤ time when this opinion was expressed: sometime in 1980¹

¹This information is not in the text.